

Neuroimaging Data Analysis Pipeline: A Framework for Clinical and Functional MRI Exploration

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Abstract

A comprehensive neuroimaging analysis pipeline was developed to integrate structural, clinical, and cognitive datasets with functional MRI processing for biomarker discovery in cognitive impairment. The framework includes automated quality control, preprocessing, group comparisons, predictive modeling, and visualization using Python. A synthetic multi-modal dataset representing 100 participants across control, mild cognitive impairment, and dementia groups was used to demonstrate reproducible workflows. Results showed group-level descriptive statistics, predictive modeling with Random Forest and Logistic Regression, and connectivity metrics from simulated 68-region fMRI time series. The pipeline outputs included Markdown reports, JSON results, and multi-panel figures suitable for neuroscientific communication.

Introduction

Neuroimaging research increasingly relies on large, multi-modal datasets combining clinical metadata, cognitive assessments, and functional or structural brain imaging. While many analytic tools exist, few end-to-end pipelines combine data ingestion, QC, statistical testing, machine learning, and automated reporting in one reproducible workflow. This study developed such a pipeline with a focus on clarity, extensibility, and publication-ready outputs. Structured data from 100 simulated participants were analyzed to identify volumetric and cognitive patterns related to diagnostic status. The framework is designed to accommodate real public datasets such as OASIS, ADNI, and the 1000 Functional Connectomes Project.

Related Work

Prior work produced key tools for neuroimaging, including FreeSurfer for cortical segmentation, FSL for fMRI analysis, and Nilearn for connectome visualization. Statistical workflows commonly use R or Python to analyze volumetric biomarkers. Machine learning in neuroimaging has applied random forests, SVMs, and deep learning to diagnosis classification from brain imaging. These components are often fragmented across specialized libraries. This project integrates all analytical stages in one modular pipeline, enabling reproducible analyses and professional documentation.

Methods

Dataset and Preparation. Participants: $n=100$ with age 18-82 years, split into Control (33), MCI (39), and Dementia (28). Features included age, education, MMSE, brain volume, hippocampal volume, and cognitive score. Synthetic data were generated to mimic OASIS-like neuroimaging registries. Functional MRI data were simulated with 68 parcellated regions and 200 timepoints.

Preprocessing and Quality Control

Quality control checked missingness and basic bounds automatically. Clinical data were encoded, z-scored, and normalized. fMRI preprocessing included connectivity computation via Pearson correlation matrices, and thresholded networks were computed at the 75th percentile to reduce spurious connections.

Statistical Analysis and Modeling

Group comparisons used independent samples t-tests with effect sizes. Correlation matrices and significance testing were computed across continuous features. Predictive models included Random Forest and Logistic Regression with stratified train-test splits.

Results

Descriptive statistics showed mean age 50.8 ± 21.1 years and mean MMSE 21.7 ± 4.4 . Control mean brain volume was 1099.1 ± 152.0 cm³ versus 1079.9 ± 124.1 cm³ for combined patients. None of the t-tests

reached $p < 0.05$, with effect sizes: brain volume -0.14, hippocampal volume 0.18, cognitive score 0.04, and MMSE -0.13. Classifier performance was modest: Random Forest AUC 0.285 and accuracy 0.533; Logistic Regression AUC 0.135 and accuracy 0.600.

Connectivity Analysis

Functional connectivity analysis across 68 regions produced mean connectivity 0.172, clustering coefficient 1.00, and global efficiency 1.00. Community detection identified 68 communities with modularity 0.040, indicating dense local connectivity in simulated networks with limited larger-scale modular structure.

Conclusion and Future Work

A modular, reproducible neuroimaging analysis pipeline was constructed and evaluated on synthetic clinical and fMRI data from 100 participants. The framework handles data generation, QC, statistical testing, predictive modeling, connectivity analysis, visualization, and report generation in one workflow. Although classification performance was modest because of synthetic labels, the architecture supports real datasets and future deep-learning extensions. Planned work includes integrating actual public neuroimaging data, EEG pipelines, and improved predictive models.

Acknowledgments

Concepts draw from OASIS, ADNI, and the 1000 Functional Connectomes Project. The project is implemented with Nilearn, NiBabel, scikit-learn, Matplotlib, Seaborn, FPDF2, and python-pptx.